

Conversion from NFA to DFA:

Converting NFA to its equivalent DFA

- In NFA, when a specific i/p is given to the current state the m/c goes to multiple states
- NFA could have zero, one or more than one on a given i/p symbol
- Where as in DFA, a specific i/p is given to the current state, the m/c goes to only one state.
- DFA has only one move on a given i/p symbol.

Let $M = (Q, \Sigma, \delta, q_0, f)$ is an NFA which accepts the lang $L(M)$. There should be equivalent DFA denoted by M'

$$M' = (Q', \Sigma', q_0', \delta', f')$$

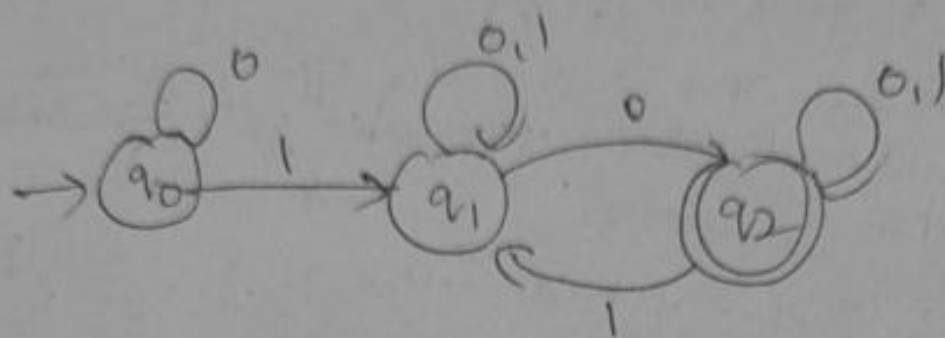
such that $L(M) = L(M')$

Steps:

- 1) initially $Q' = \emptyset$
- 2) Add q_0 of NFA to Q' . Then find transitions from this start state.
- 3) In Q' find the possible set of states for each i/p symbol. If this set of states is not in Q' , then added to Q' .

Step 4:- In DFA the final state will be all the states which contain F (final states of NFA)

Q) Convert the given NFA to DFA



Transition table

	0	1
q_0	q_0	q_1
q_1	$\{q_1, q_2\}$	q_1
q_2	q_2	$\{q_1, q_2\}$

Now obtain δ' transition from q_0

$$Q' = \{q_0\}$$

$$\delta'([q_0], 0) = [q_0]$$

$$\delta'([q_0], 1) = [q_1]$$

now add $Q' = \{q_0, q_1\}$

$$\delta'([q_1], 0) = [q_1, q_2] \quad (\text{new state generated})$$

$$\delta'([q_1], 1) = [q_1]$$

$$Q' = \{q_0, q_1, [q_1, q_2]\}$$

the δ' transition for q_2

$$\delta'([q_2], 0) = [q_2]$$

$$\delta'([q_2], 1) = [q_1, q_2]$$

$$Q' = \{q_0, q_1, q_2, [q_1, q_2]\}$$

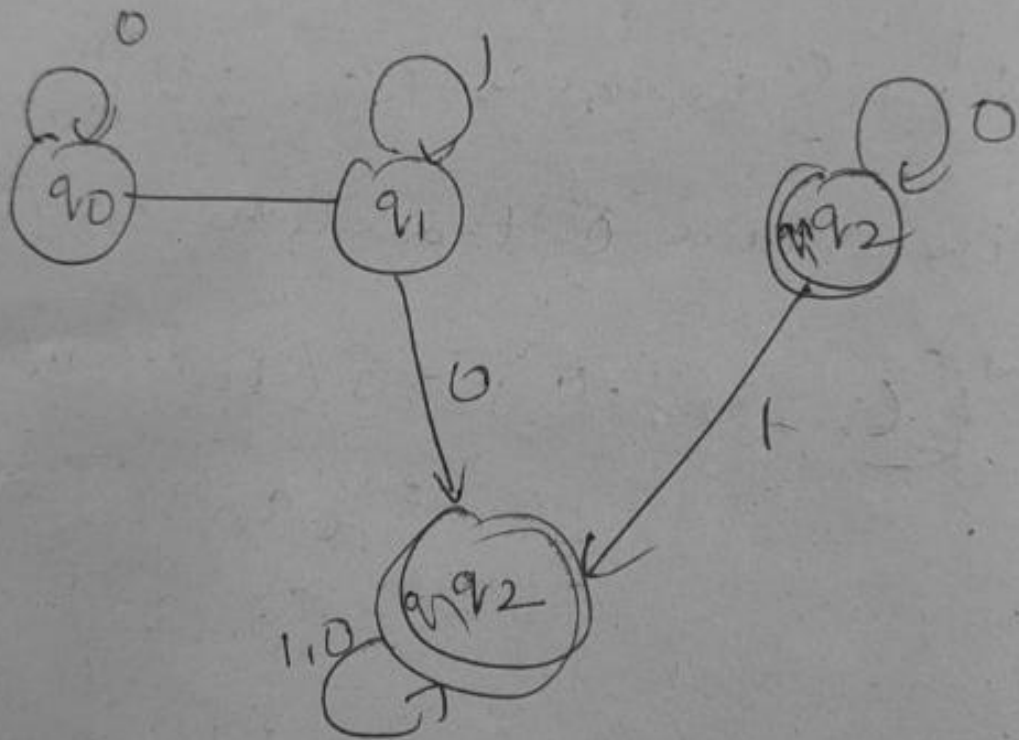
Now obtain δ' transition on $[q_1, q_2]$

$$\begin{aligned} \delta'([q_1, q_2], 0) &= \delta(q_1, 0) \cup \delta(q_2, 0) \\ &= [q_1, q_2] \cup [q_2] \\ &= [q_1, q_2] \end{aligned}$$

$$\begin{aligned} \delta'([q_1, q_2], 1) &= \delta(q_1, 1) \cup \delta(q_2, 1) \\ &= [q_1] \cup [q_1, q_2] \\ &= [q_1, q_2] \end{aligned}$$

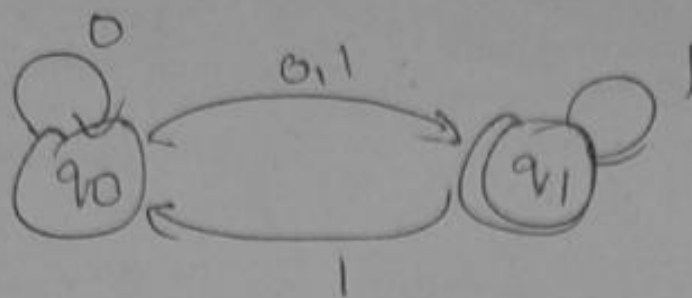
The transition table for DFA is

	0	1
q_0	q_0	q_1
q_1	q_1, q_2	q_1
q_2	q_2	q_1, q_2
q_1, q_2	q_1, q_2	q_1, q_2



now the state q_2 in this case is an Unreachable state. Hence it can be eliminated.

Q) Convert the given NFA to DFA



Solⁿ

Transition table

	0	1
q ₀	q ₀ q ₁	q ₁
q ₁	—	q ₀ q ₁

δ' transitions for q₀

$$\delta'(q_0, 0) = q_0q_1 \text{ (new state)}$$

$$\delta'(q_0, 1) = q_1$$

δ' transitions for q₁

$$\delta'(q_1, 0) = \phi$$

$$\delta'(q_1, 1) = q_0q_1$$

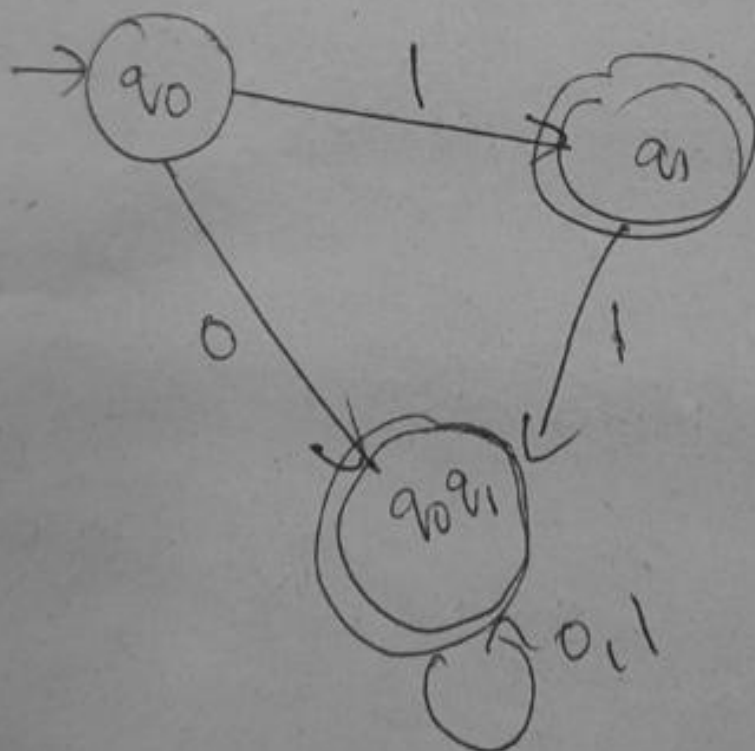
δ' transitions for q₀q₁

$$\delta'(q_0q_1, 0) = \delta(q_0, 0) \cup \delta(q_1, 0) = q_0q_1$$

$$\delta'(q_0q_1, 1) = q_0q_1$$

Transition table for DFA is

	0	1
q_0	q_0q_1	q_1
q_1	$\emptyset$	q_0q_1
q_0q_1	q_0q_1	q_0q_1

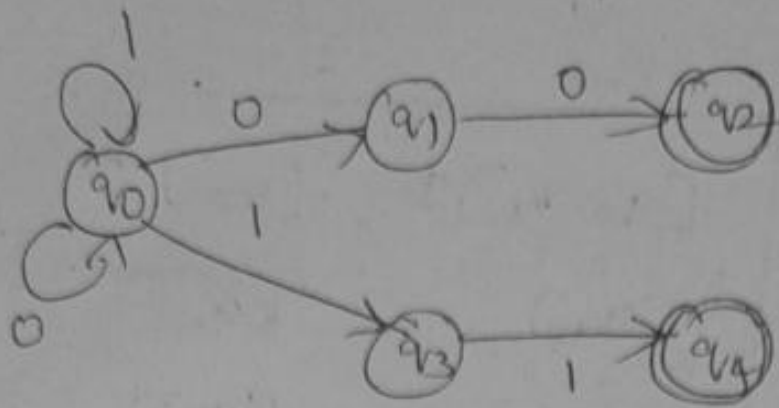


a) Convert the given NFA to DFA

state \ i/p	0	1
q_0	$\{q_0, q_1\}$	q_0
q_1	q_2	q_1
q_2	q_3	q_3
q_3	ϕ	q_2

State	0	1
$[q_0] A$	$[q_0 q_1] B$	$[q_0] A$
$[q_1]$	$[q_2]$	$[q_1]$
$[q_2]$	$[q_3]$	$[q_3]$
$[q_3]$	ϕ	$[q_2]$
$[q_0 q_1] B$	$[q_0, q_1, q_2] C$	$[q_0, q_1] B^{\oplus}$
$[q_0, q_1, q_2] C$	$[q_0, q_1, q_2, q_3] D$	$[q_0, q_1, q_3] E$
$[q_0, q_1, q_3] E$	$[q_0, q_1, q_2] C$	$[q_0, q_1, q_2] C$
$[q_0, q_1, q_2, q_3] D$	$[q_0, q_1, q_2, q_3] D$	$[q_0, q_1, q_2, q_3] D$

Convert NFA to its equivalent DFA



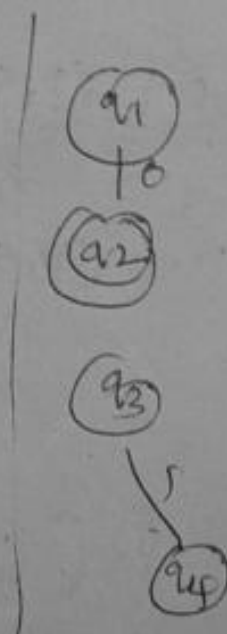
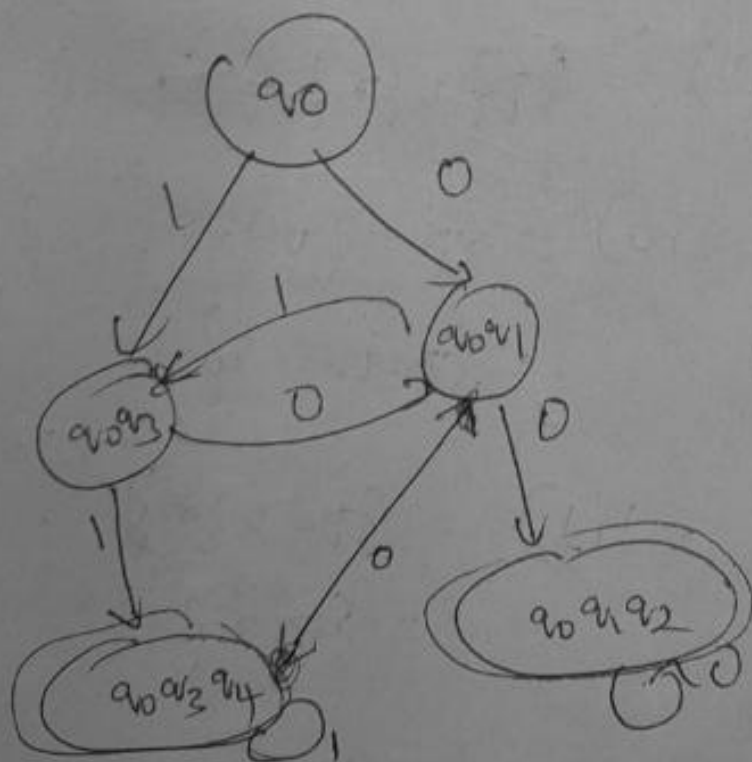
	0	1
q_0	$\{q_0, q_1\}$	$\{q_0, q_3\}$
q_1	$\{q_2\}$	ϕ
q_2	ϕ	ϕ
q_3	ϕ	$\{q_4\}$
q_4	ϕ	ϕ

$$\delta'([q_0], 0) = \{q_0, q_1\}$$

$$\delta'([q_0, q_1], 0) = \{q_0, q_1, q_2\}$$

$$\begin{aligned} \delta'([q_0, q_1], 1) &= \{q_0, q_3\} \cup \{\phi\} \\ &= \{q_0, q_3\} \end{aligned}$$

	0	1
$[q_0]$	$[q_0 q_1]$	$[q_0 q_3]$
$[q_1]$	q_2	ϕ
q_2	ϕ	ϕ
q_3	ϕ	q_4
q_4	ϕ	ϕ
$q_0 q_1$	$q_0 q_1 q_2$	$q_0 q_3$
$q_0 q_3$	$q_0 q_1$	$q_0 q_3 q_4$
$q_0 q_1 q_2$	$q_0 q_1 q_2$	$q_0 q_3$
$q_0 q_3 q_4$	$q_0 q_1$	$q_0 q_3 q_4$



Q: NFA equivalent to DFA

	0	1
P	$\{P, Q\}$	P
Q	r	r
r	s	-
s	s	s

$$\delta^*(CP, \alpha, s), 1) = \{P, \alpha, s\}$$

	0	1
$[P]$	$[CP, \alpha]$	P
$[\alpha]$	$[CP, \alpha]$	α
$[s]$	$[s]$	$[s]$
$[CP, \alpha]$	$[CP, \alpha, \alpha]$	$[CP, \alpha]$
$[CP, \alpha, \alpha]$	$[P, \alpha, \alpha, s]$	$[P, \alpha]$
$[P, \alpha]$	$[P, \alpha, s]$	$[P]$
$[P, \alpha, s]$	$[CP, \alpha, \alpha, s]$	$[CP, \alpha, s]$
$[P, \alpha, s]$	$[CP, \alpha, \alpha, s]$	$[P, \alpha, s]$
$[CP, \alpha, s]$	$[CP, \alpha, s]$	$[CP, s]$
$[CP, s]$	$[CP, \alpha, s]$	$[CP, s]$

